Technical Writing
Mid Summary

هذا الملخص تم كتابته من قبل الطالب : ليث عمرو

هذا التلخيص هو جهد بشري قابل للخطأ،
إن أصبت فمن الله، وإن أخطأت فمنّي ومن الشيطان.

هذا العمل مقدّم ثوابه لروح والدي (قاسم مفيد عمرو)
فمن وجدَ فيه فائدة فليدعُ له بالرحمة والمغفرة.

هذا الملخص لا يغني بتاتًا عن دراسة الكتاب وهو فقط للمراجعة السريعة قبل الاختبار.

Technical Writing
Mid Summary

Section One

1. Definition and Purpose of Technical Writing

- Technical writing is an act of **communication**, not self-expression.
- The focus is on **presenting information clearly**, not the writer's attitude.
- It is the type of writing professionals use in their jobs.

2. Communication vs. Self-Expression

- **College writing**: Creative, expository, and self-expressive (e.g., essays, poems).
- **On-the-job writing**: Technical, instrumental, and communicative (e.g., reports, memos).

3. The Communication Triangle

- **Technical writing** emphasizes the link between the **reader and the subject**. The writer's role is secondary.
- **Creative writing** highlights the connection between the **writer and the subject** (e.g., poems).
- **Expository writing** focuses on the relationship between the **writer and the reader** (e.g., essays).

4. Major Types of On-the-Job Writing

1. **Correspondence**:
    - **Memos**: Internal documents for brief exchanges (e.g., interoffice memos).
    - **Letterhead Letters**: Formal external communications (e.g., business letters).
    - **Nonletterhead Letters**: Informal external communications (e.g., customer inquiries).
2. **External Publications:**
    - Sales, marketing, and promotional materials.
    - Manuals, handbooks, and research reports.
    - Corporate financial reports.
3. **Internal Documents:**
    - Routine project reports.
    - Management decision-making documents.
    - Policy/procedure manuals and archival data.

Technical Writing
Mid Summary

5. The People Who Write

- Supervisors, managers, and executives.
- Technical experts and researchers.
- Technicians and professional writers/editors.

Technical Writing
Mid Summary

Section Two

1. The Continuum of Writing Quality

Technical writing quality exists on a spectrum:

- **Unacceptable** (0–20%): Hinders professional growth, requires revisions.
- **Adequate** (20–80%): Meets basic standards but lacks distinction.
- **Excellent** (80–100%): Gains recognition, drives success.

2. Categories of Technical Writing

1. **Excellent Technical Writing:**
    - Leads to promotions, funded proposals, and praise.
    - Stands out in clarity, impact, and memorability.
2. **Adequate Technical Writing:**
    - Avoid embarrassment; accepted without comment.
    - Neither harms nor significantly benefits the writer.
3. **Unacceptable Technical Writing:**
    - Requires rework, limits career advancement.
    - Reflects poorly on other technical skills.

3. Criteria for Adequacy

A document is **adequate** if it is:

- **Useful:** Addresses the problem and solution.
- **Accurate:** Reliable and valid information.
- **Complete:** No missing data or steps.
- **Orderly:** Follows organizational conventions.
- **Correct:** Free of grammatical/mechanical errors.

Checklist for Adequacy:

- **Format:** Proper margins, headings, figures, references.
- **Organization:** Logical structure, balanced sections.
- **Usage/Mechanics:** Error-free spelling, punctuation, grammar.
- **Style:** Direct, concise, problem focused.

Technical Writing
Mid Summary

4. Levels of Adequacy

An **adequate** document may also be:

- Brief, smooth, and direct.
- More than adequate in some respects.

5. Criteria for Excellence

Beyond adequacy, **excellent** writing is:

- Easy and interesting to read.
- Memorable.
- Brief, smooth, and direct.

6. Demands for Excellence

Excellence is critical when:

- Addressing **large/important audiences**.
- Involving **safety, money, or reputations**.

7. Achieving Adequacy and Excellence

- **Key principles:**
  1. **Adequacy is achievable by anyone** following rules.
  2. **Excellence requires iteration**: Improve adequate drafts.
  3. **Excellence demands creativity**: Experiment with new ideas.
- **Steps to Excellence:**
  - Start with an **adequate** draft.
  - Analyze personal writing strengths/weaknesses.
  - Refine paragraphs, sentences, and word choice.
  - Use cohesive devices (transitions, logical flow).

8. Final Reminders

- **Do not expect immediate excellence**—build gradually.
- **Revise iteratively** to elevate adequate work.
- **Focus on foundational skills** first.

Technical Writing
Mid Summary

Section Three

1. Importance of Context Analysis

Technical writing requires understanding:

- **Audience** (Who will read it?).
- **Purpose** (Why are you writing it?).
- **Type of information** (What is being communicated?).

Key questions: **WHO?, WHY?, WHAT?**

2. Varieties of Audience

- **Audiences differ in expertise and needs:**
  1. **Executives/Administrators**: Focus on policy and decision-making.
  2. **Technical Experts**: Require detailed, specialized information.
  3. **Technicians**: Need practical, actionable instructions.
  4. **Laypeople**: Lack expertise; require simplified explanations.
  5. **Combined Audience**: Mix of the above (e.g., a report for both managers and technicians).
- **Types of Audiences in Organizations**:
  - **Primary**: Directly affected by the information.
  - **Secondary**: Indirectly affected.
  - **Immediate**: The person receiving/documenting the report (e.g., a supervisor).
- **Flow of Information**:
  - Top executives → Boss → Co-workers → Subordinates.
  - External stakeholders (e.g., counterparts in other organizations).

Technical Writing
Mid Summary

3. Purpose of Communication

- **Primary Purpose**: Main goal for the major audience (e.g., convincing executives to approve a proposal).
- **Secondary Purposes**: Additional goals (e.g., demonstrating expertise to technical peers).
- **Personal Goals**: Align with organizational objectives (e.g., career advancement).

Example: A recommendation report may aim to:

- Persuade decision-makers.
- Validate technical credibility.
- Save time for busy executives.

4. Types of Technical Information

Guidelines for effective content:

- **Avoid information dumping:** Separate topics into distinct reports.
- **Relevance:** Exclude details readers already know, don't need, or don't care about.
- **Focus:** Stick to the point; avoid combining unrelated projects/favors in one document.
- **Temporal Separation:** Distinguish transient data from long-term information.

5. Report Design Worksheet

A tool to document context analysis:

- **Preliminary Information:** Task, deadlines, format restrictions.
- **Reader Analysis:**
    - Audience roles (primary/secondary/immediate).
    - Expertise, attitudes, and potential impact of the report.
- **Reader's Purpose:** What they should know/do after reading.
- **Writer's Purpose:** Desired outcomes (organizational/personal).

Technical Writing
Mid Summary

6. Developing Design Criteria

- **Key parameters to tailor reports:**
  1. **Audience**: Adjust language, depth, and tone.
  2. **Purpose**: Align content with goals (e.g., persuasion vs. instruction).
  3. **Type of Information**: Structure based on relevance and complexity.
  4. **Vectors of Information Flow**: Consider organizational hierarchy.
- **Resulting Design Choices**:
  - Report type (e.g., memo vs. formal report).
  - Content distribution (main body vs. appendices).
  - Language level (technical vs. layman's terms).
  - Writing style, graphics, and revision intensity.
  - Copy distribution (who receives the report).

Technical Writing
Mid Summary

Section Four

1. Importance of Correctness

- Technical writing must be **free from errors** in grammar, mechanics, and usage.
- Ensures clarity, professionalism, and credibility.

2. Editing for Correctness

- Steps to Improve Editing:

1. **Acknowledge weaknesses** in grammar/mechanics.
2. **Allocate extra time** for proofreading.
3. **List recurring errors** and use them as an editing checklist.
4. **Read backward** (sentence by sentence) to spot mistakes.
5. **Seek informal reviews** from peers.
6. **Recheck corrections** after revisions.
7. **Take breaks** before final proofreading for fresh perspective.

3. Common Grammar Problems

1. **Subject-Verb Agreement**:
   - Singular subjects → singular verbs; plural subjects → plural verbs.
   - *Example*: "Jack and Jill **are** coming" (not "is").
2. **Pronoun-Antecedent Agreement**:
   - Pronouns must match their antecedents in number/gender.
   - *Example*: "Each soldier must clean **his** rifle" (not "their").
3. **Grammatical Parallelism**:
   - Items in lists/headings must follow the same structure.
   - *Example*: "Sort the cards. Load the deck. Clear the jam." (all imperative).
4. **Sentence Fragments**:
   - Incomplete sentences must be revised.
   - *Example*: "Harold went to the store **and** the drugstore." (not two separate fragments).

Technical Writing
Mid Summary

4. Common Punctuation Problems

- **End Punctuation:** Use periods (.), question marks (?), or exclamation marks (!) correctly.
  - *Example*: "Can you finish by noon?" (not a period).
- **Introductory Clauses:** Separate with commas.
  - *Example*: "Because she was happy, Helen smiled."
- **Interruptive Elements:** Use commas, dashes, or parentheses.
  - *Example*: "Jack (the man in front) gave me tickets."
- **Possessive Apostrophes:**
  - *Example*: "John's hat" (not "Johns hat").
- **Lists:** Use colons (:) to introduce lists.
  - *Example*: "Components include: CPU, memory, and I/O devices."
- **Quotation Marks:** Place punctuation inside quotes.
  - *Example*: "River of the woods." (not outside).

5. Conventions in Mechanics

1. **Abbreviations/Symbols:**
   - Use standard symbols ($, %, °C, & in company names).
   - Common abbreviations: FBI, WHO, etc., e.g., rpm.
2. **Numbers/Numerals:**
   - Spell out 1–9; use numerals for 10+.
   - Exceptions:
     - Addresses, dates, money (*March 9, $5.00*).
     - Mixed numbers (*3½*), decimals (*0.5*).
     - Hyphenate dimensions (*55-gallon barrel*).
3. **Italics/Underlining:**
   - Titles of books/journals (*Journal of Science*).
   - Latin species names (*Homo sapiens*).
   - Foreign words not in common use (*résumé*).

Technical Writing
Mid Summary

Key Takeaways

- **Editing is systematic:** Use checklists, backward reading, and peer reviews.
- **Grammar/Punctuation:** Master agreement, parallelism, and sentence structure.
- **Mechanics:** Follow conventions for abbreviations, numbers, and formatting.

Technical Writing
Mid Summary

Section Five

1. Definition and Classification of Reports

- **Broad Definition:** Includes proposals, literature reviews, manuals, environmental impact statements, and journal articles.
- **Classification Criteria:** Based on **form** (structure) and **function** (purpose).

2. Variations in Report Form

1. **Fill-in-the-Blank Reports:**
    - Pre-formatted templates for repetitive tasks (e.g., budget summaries, inspection reports).
    - Focus on numerical data.
    - *Example*: Monthly financial reports with predefined categories.
2. **Letters and Memos:**
    - Short (1–5 pages), informal.
    - May include headings or tables but often lack formal structure.
    - *Example*: Interoffice memos or recommendation letters.
3. **Formal Reports:**
    - Structured with **front matter** and **back matter**.
    - Used for comprehensive documentation.

3. Components of Formal Reports

A. Front Matter:

1. **Cover Letter/Memo:** Introduces the report to the audience.
2. **Cover (Optional):** Protects and identifies the report (title, publisher, design).
3. **Title Page:** Includes title, author, organization, date, and bibliographic data.
4. **Table of Contents:** For reports ≥15 pages; lists chapters/sections.
5. **List of Illustrations:** If figures/tables are frequently referenced.
6. **List of Definitions/Abbreviations:** Clarifies specialized terms.
7. **Summary/Abstract:** Concise overview for busy executives.

B. Back Matter:

1. **Bibliography/References:** Lists cited sources.
2. **Appendixes:** Supplemental data (calculations, raw data, resumes).
3. **Index (Optional):** For manuals/long reference documents.

C. Page Numbering:

- Front matter: Roman numerals (i, ii, iii).
- Main text: Starts with page 1.
- Back matter: Continues numbering or uses lettered appendices (A-1, A-2).

4. Semi-Formal Reports

- Simplified versions of formal reports (e.g., missing bibliography or appendixes).
- Most common in workplace settings.

5. Variations in Report Function

Reports are categorized by their **purpose**:

1. **Project Reports:**
    - Track progress, status, and future needs.
    - *Examples*: Progress reports, quarterly updates.
2. **Activities Reports:**
    - Document time spent on tasks (e.g., travel, training).
    - Used for performance evaluation.
    - *Examples*: Monthly activity logs, accident reports.
3. **Reference Data Reports:**
    - Present collected data for others' use.
    - *Example*: Journal articles.
4. **Management Reports:**
    - Support decision-making (e.g., proposals, feasibility studies).
    - Include project/activity reports and recommendations.

Technical Writing
Mid Summary

Key Takeaways

- **Form Follows Function:** Choose the report type (fill-in, memo, formal) based on audience and purpose.
- **Formal Reports:** Require structured front/back matter for clarity and professionalism.
- **Workplace Reports:** Most are semi-formal or functional (project/activity/management-focused).

Technical Writing
Mid Summary

Section Six

1. Five Phases of Report Writing (Two of them before start writing)

1. **Data Gathering**
2. **Preliminary Organization**
3. **Initial Draft Preparation**
4. **Revision**
5. **Final Inspection**

2. Data Gathering Methods

Four primary approaches to collect information:

1. **Direct Observation**: Firsthand data collection.
2. **Testing and Experimentation**: Empirical data through experiments.
3. **Consultation and Interview**: Insights from experts or stakeholders.
4. **Library Research**: Secondary data from published sources.

3. Preliminary Organization

Goal: Ensure readers can easily locate needed information.
Principles of Good Organization:

- **Group related items** together.
- **Separate unrelated items**.
- **Label sections clearly**.

Steps to Achieve Sound Organization:

1. Analyze the communication context (audience, purpose).
2. List potential report content.
3. Cluster related information into "shoeboxes" (categories).
4. Label categories and create subcategories.

Technical Writing
Mid Summary

4. Initial Draft Preparation

- Key Guidelines:

- **Focus on content,** not style or grammar.
- **Start with the easiest section** (e.g., methods/results).
- **Delay writing the introduction** (hardest part).
- **Write quickly** without self-editing.

5. Revision

Importance: Distinguishes strong writers from weak ones.

- Process:

- **Take a break** (1–2 days) before revising.
- **Major Revision Areas:**
  - Reorganize sections for clarity.
  - Restructure sentences (grammar, word choice).
  - Add/delete large sections as needed.
  - Refine graphics and illustrations.
  - Polish writing style.

- Revision Checklist:

- Grammar, spelling, punctuation.
- Consistency in abbreviations, numbers, terminology.
- Proper headings, captions, and page numbering.

6. Final Inspection

Purpose: Catch errors introduced during final drafting or overlooked earlier.

- Focus Areas:

- Typographical errors.
- Misspelled names or numerical data.
- Missing/repeated words or pages.
- Inconsistent abbreviations or symbols.
- Common mistakes (e.g., "its'" vs. "its").

Technical Writing
Mid Summary

Section Seven

1. Importance of Graphics in Technical Writing

- Graphics (tables, charts, diagrams, photos) complement verbal communication.
- They:
    - Clarify complex information.
    - Summarize numerical data efficiently.
    - Enhance reader understanding.

2. Key Terminology

1. **Table:** Organized columns of precise (often numerical) data.
2. **Figure:** Includes charts, graphs, line drawings, and photographs.
3. **Caption:** Figure/table number + brief description (e.g., *Table 15-2: Evaporator Correction Factors*).
4. **Reference:** In-text pointer to a graphic (e.g., "See Figure 5").
5. **Callout:** Labels identifying parts of an illustration (verbal or numerical).
6. **Legend:** Explains symbols/abbreviations used in graphics.

3. Types of Illustrations

A. Photographs vs. Line Drawings

- **Photographs:**
    - Used in manuals, brochures.
    - Show real-world appearance.
- **Line Drawings:**
    - Remove unnecessary details.
    - Highlight critical components.

Technical Writing
Mid Summary

B. Line Drawing Varieties

1. **2D Drawing:** Standard mechanical drafting (plane view).
2. **3D Drawing:** Realistic representation for easier visualization.
3. **Cross-Section:** Reveals internal structure as if sliced.
4. **Exploded View:** Disassembles parts along an axis to show relationships.
5. **Cutaway View:** Removes external housing to expose internal mechanisms.

4. Examples and Applications

- **Tables:** Present precise data (e.g., correction factors, specifications).
- **Figures:**
    - *Flowcharts*: Process steps.
    - *Graphs*: Trends/comparisons.
    - *Diagrams*: Assembly instructions (e.g., exploded views for repair manuals).
- **Callouts:** Direct attention (e.g., "Solenoid Valve" in a diagram).

5. Best Practices

- **Clarity:** Use simple, labeled visuals.
- **Relevance:** Ensure graphics directly support the text.
- **Consistency:** Maintain uniform styling (fonts, line weights).
- **Accessibility:** Provide captions/legends for all graphics.

تم بفضل الله،

اللهمّ ارحم (قاسم مفيد عمرو) واجعله من ورثة جنّة النعيم.